

Sentinel AML — Blueprint & Product Requirements Document

AI-Assisted AML Detection & Investigation Platform Capstone Project — Leadership with AI, IIT Mumbai

Prepared by: [Team names — fill in] Date: 27 September 2026 Status: Draft — open items pending team confirmation (see Section 16)

All customer, account, and transaction data referenced anywhere in this document is fictional, created for demonstration purposes only. This document does not represent a production AML compliance system.

Table of Contents

- 1. Project Vision & Business Case
- 2. Product Requirements Document (PRD)
- 3. User Personas & Stakeholders
- 4. Current Process / Workflow
- 5. Functional Requirements & Use-Case Document
- 6. Technical Architecture
- 7. Data Requirements
- 8. Integration / API Specifications
- 9. UI Wireframes / Screen Designs
- 10. Non-Functional Requirements
- 11. Acceptance Criteria / Definition of Done
- 12. Flowchart 1 – End-to-End Data Flow
- 13. Flowchart 2 – Investigation Process Flow
- 14. Risks, Assumptions, Dependencies & Constraints
- 15. Delivery Plan / Milestone Roadmap
- 16. Meeting Notes and Decisions (Open Items Log)

1. Project Vision & Business Case

Product: Sentinel AML — AI-Assisted AML Detection & Investigation Platform **Tagline:** From transaction to trigger to traceable decision. **Document owner:** [assign] **Status:** Draft for team review — capstone academic project, fictional data only

1. Purpose

Sentinel AML is an end-to-end product, not a chatbot. It watches a bank’s transaction activity, automatically **detects** patterns that warrant investigation, and gives an AML investigator an **AI-assisted workspace** to investigate the case — evidence gathered, connections traced, red flags identified, and a structured report drafted — while the human investigator retains every decision.

It has three moving parts a reader should picture immediately:

- 1. **A trigger.** Something happens in the transaction feed that the system is watching for.
- 2. **A landing experience.** The investigator opens the product and sees a queue of cases, not a blank chat box.
- 3. **An investigation.** The AI assembles the evidence; the human decides.

2. Business Problem

Financial institutions must review transaction-monitoring alerts for possible money laundering. Today this review is manual: an investigator opens 6–8 separate systems (core banking, KYC, case management, screening tools) to piece together the customer’s history, trace where money went, check for related parties, and write up findings. This is slow, inconsistent between investigators, and hard to audit after the fact.

Two consequences of the status quo:

- **Investigation backlog and inconsistent quality** — investigators spend most of their time compiling evidence rather than applying judgment.
- **Weak traceability** — when a regulator or auditor asks “how was this decided,” the answer is scattered across systems and inboxes.

3. Business Case (why build this)

Driver	Current state (assumption, to validate with real data if this goes beyond the capstone)	Target state
Time per alert	Investigator manually compiles evidence across systems — assumed several hours per moderately complex case	AI assembles the evidence pack in minutes; investigator spends time on judgment, not compilation
Consistency	Write-ups vary by investigator	Structured report format every time, same evidence categories
Auditability	Evidence trail reconstructed after the fact from emails/notes	Every AI action and human decision logged automatically as it happens
Missed connections	Related accounts / prior alerts easy to miss under time pressure	System explicitly traces fund flow and relationships, and lists what is missing

All time/effort figures above are illustrative assumptions for the capstone, not measured data, and are flagged as such throughout this pack (see **Section 14 — Risks, Assumptions, Dependencies & Constraints**).

4. Vision Statement

Give every AML investigator a case that already has its evidence gathered, its connections traced, and its open questions written down — so their first hour on a case is spent deciding, not searching.

5. Who This Is For

See **Section 3 — User Personas & Stakeholders** for full detail. In short: AML Investigators (primary users), Investigation Team Leads, Compliance Officers, and a System Admin role for rule/threshold configuration.

6. Scope of This Capstone

In scope

- A rule-based **trigger engine** that scans a mock transaction feed and raises alerts (no AI in this layer — deliberately rules-based; see **Section 6 — Technical Architecture**).
- A **case queue landing page** (the product’s home screen after login).
- An **AI-assisted investigation workspace** for one frozen scenario family: *sudden high-value credit followed by rapid movement of funds*, demonstrated with a suspicious case and a legitimate “twin” case (see **Section 5 — Functional Requirements & Use-Case Document**).
- A **human decision layer**, kept visually and structurally separate from AI output.
- A **light escalation queue** for a Compliance role.
- A **full audit log** of AI actions and human decisions.
- All data is **fictional / mock**, clearly labelled throughout the product and documentation.

Out of scope

- Real integrations with core banking, KYC providers, credit/sanctions bureaus, or case-management systems.
- Automated regulatory filing (STR/CTR) or any AI-made legal/criminal determination.
- Full identity and access management, production-grade security certification, multi-jurisdiction support.
- Every AML typology — only the frozen scenario family above, plus documented extension points.

7. Success Measures (for the capstone demo, not a production KPI set)

Measure	Target for demo
End-to-end story is demonstrable live	Trigger fires → case appears in queue → AI report generated → investigator decision recorded → audit entry visible
Twin-case distinction is clear	Suspicious case (C1) and legitimate case (C2) produce visibly different AI findings from the same trigger rule
Every AI claim is traceable	100% of AI findings cite an underlying mock record or are labelled Missing/Inferred
Human boundary is visible	No screen implies the AI decided the case; every disposition requires a human action + rationale

Production-style KPIs (turnaround time, false-positive rate, adoption) belong in the roadmap document (**Section 15 — Delivery Plan / Milestone Roadmap**) as **forward-looking targets**, not measured results.

8. Budget & Timeline Assumptions

This is an **academic capstone**, not a funded product build.

- **Team:** 7 members (see **Section 3 — User Personas & Stakeholders** for the team roster and role split, and the team plan workbook for the task-level split).

- **Timeline:** submission window is the 1st–2nd week of October 2026; working backward, internal freeze target is ~6 October 2026 (adjust once the exact date is confirmed).
- **Budget:** none — uses free/low-cost LLM API tier, open-source libraries, mock data. Any live LLM API cost is the only real spend, kept small by caching demo responses (see **Section 6 — Technical Architecture**).
- **Effort assumption:** a working prototype covering the in-scope items above, not a production system.

9. What “Done” Looks Like for This Capstone

A submitted blueprint document, a slide deck, and a running (or reliably screen-recorded) prototype that together show: a transaction pattern gets detected → an investigator opens the case in a real product screen → the AI has already assembled evidence and flagged what’s missing → the investigator makes and records the decision → the whole thing is logged. See **Section 11 — Acceptance Criteria / Definition of Done** for the checklist version.

2. Product Requirements Document (PRD)

Product: Sentinel AML **Related:** **Section 1 — Project Vision & Business Case, Section 5 — Functional Requirements & Use-Case Document**

1. Product Overview

Sentinel AML has two halves that together make it a product rather than a single AI feature:

1. **Detect** — a rule-based Trigger Engine continuously scans transaction activity and raises alerts when defined conditions are met.
2. **Investigate** — an AI-assisted workspace turns each alert into a structured, evidence-linked case for a human investigator to review and decide.

2. Features

ID	Feature	Description	Priority
F1	Login / role landing	Role-based entry (Investigator, Team Lead, Compliance, Admin); mock auth for prototype	Must
F2	Case queue dashboard	Landing page after login: open cases, priority, alert type, SLA/age, quick filters	Must
F3	Trigger engine	Background rule evaluation over mock transaction feed; generates alerts with rule ID + threshold values shown	Must
F4	Case intake	Alert becomes a case; claimed by an investigator	Must
F5	Investigation workspace	Case header, alert summary, transaction timeline, fund-flow/relationship view, AI findings panel, investigation questions, recommended next steps	Must
F6	Evidence status tagging	Every AI finding tagged Verified / Inferred / Missing / Conflicting with citations	Must
F7	Human decision panel	Separate UI region; actions = request info / add evidence / escalate / close; rationale mandatory	Must
F8	Audit log	Immutable, per-case and system-wide log of AI actions and human decisions	Must
F9	Escalation / compliance queue	Escalated cases appear in a Compliance role view	Should
F10	Admin rule configuration (light)	View/adjust the two trigger rule thresholds; view/edit playbook	Should
F11	Twin-case scenario library	Pre-loaded suspicious case (C1) and legitimate case (C2) sharing the same trigger, for demo and testing	Must
F12	Cached / live report toggle	AI report can replay from cache (demo reliability) or regenerate live	Should
F13	Second trigger scenario (velocity/structuring)	Additional rule + case for variety	Could
F14	Network graph visualization	Visual account/fund-flow graph in the workspace	Could

MoSCoW note: **Must** = required for capstone demo/acceptance; **Should** = strengthens the demo, build if time allows; **Could** = stretch, defer to roadmap if short on time. See **Section 15 — Delivery Plan / Milestone Roadmap**.

3. Business Rules

#	Rule
BR1	The AI never sets, changes, or recommends a final legal/criminal determination. It surfaces indicators and evidence status only.
BR2	Every AI finding must cite the record(s) it is based on. A finding with no citation is not permitted in the output.
BR3	“Verified” may only be used when the underlying data directly supports the claim; anything inferred from multiple data points is “Inferred,” never “Verified.”
BR4	Case disposition (close/escalate) is a human action only, and requires a written rationale before it can be recorded.
BR5	The Trigger Engine is rule-based only — no AI/LLM involvement in raising an alert (keeps detection deterministic and explainable).
BR6	AI tools are read-only against the mock data store. The AI cannot alter alert status, transaction records, or KYC data.
BR7	Recommended next steps must cite a playbook entry (mock internal procedure), never be generated from the model’s unstated assumptions about regulation.

#	Rule
BR8	All data, customers, accounts, and transactions are fictional; the product must visibly label itself as a prototype using mock data on every screen.

4. Out-of-Scope Items (explicit)

- Real-time payment blocking or transaction interdiction.
- Regulatory report (STR/CTR) generation or submission.
- Full KYC onboarding / remediation workflow.
- Automatic case closure without human action.
- Any claim that the product performs a legal or definitive AML determination.
- Production identity/access management, encryption-at-rest infrastructure, and security certification (documented as target-state requirements only — see **Section 10 — Non-Functional Requirements**).

5. Assumptions

- One institution, one jurisdiction context (India) for terminology and illustrative scenario framing; no jurisdiction-specific legal claims are made (see **Section 14 — Risks, Assumptions, Dependencies & Constraints**).
- A single LLM provider with tool-calling and JSON-schema output is available to the team (provider/key TBD by team — open item, see Meeting Notes).
- Prototype runs locally / on a shared demo machine; no production hosting required.

6. Dependencies

- Mock dataset (customers, accounts, transactions, relationships, alerts, documents, playbook) — see **Section 7 — Data Requirements**.
- LLM API access — see **Section 8 — Integration / API Specifications**.

7. Release Criteria

See **Section 11 — Acceptance Criteria / Definition of Done**.

3. User Personas & Stakeholders

Product: Sentinel AML

1. Personas (who uses the product)

Persona 1 — AML Investigator (primary user)

- **Role in product:** Reviews assigned cases in the investigation workspace, works through evidence, makes and records decisions.
- **Goals:** Close/escalate cases accurately and quickly; avoid missing real red flags; avoid over-flagging legitimate activity.
- **Pain today:** Compiling evidence across systems is slow; hard to be sure nothing was missed; write-ups are inconsistent across the team.
- **What Sentinel AML gives them:** A pre-assembled, cited evidence pack, explicit “missing information” list, and a structured decision panel with mandatory rationale.
- **Primary screens:** Case queue, Investigation workspace, Human decision panel.

Persona 2 — Investigation Team Lead

- **Role in product:** Monitors team caseload, reassigns cases, reviews quality on a sample basis.
- **Goals:** Balance workload, meet SLA, spot inconsistent decisions.
- **Primary screens:** Case queue (team view), Audit log.

Persona 3 — Compliance Officer (MLRO-delegate view)

- **Role in product:** Reviews escalated cases only; does not use the AI investigation workspace directly for routine cases.
- **Goals:** Confirm escalation is warranted; decide on further action (outside product scope — regulatory filing is manual/out of scope).
- **Primary screens:** Escalation / compliance queue, Audit log, case read view.

Persona 4 — System Admin (light role)

- **Role in product:** Configures the two trigger-engine rule thresholds; manages the mock playbook; manages user role assignment (mock auth).
- **Goals:** Keep detection rules current; keep next-step guidance (playbook) accurate.
- **Primary screens:** Admin / rule configuration.

Persona 5 — Capstone Presenter / Demo Viewer (prototype-only persona)

- **Role in product:** Not a real product persona — represents the evaluator watching the 15-minute demo.
- **Goals:** Quickly understand what triggers detection, how AI helps, and where the human decides.
- **Implication for design:** The product must be legible to a first-time viewer within minutes — favors a visible trigger moment and a clearly separated AI vs. human panel (see **Section 9 — UI Wireframes / Screen Designs**).

2. Stakeholder List

Stakeholder	Interest	Involvement
Capstone team (7 members)	Build, present, and be evaluated on the product	Build/approve everything in this pack
IIT Mumbai faculty / evaluation panel	Assess against the rubric (problem relevance, process understanding, data-driven design, AI appropriateness, architecture, guardrails, roadmap, value logic, communication)	Approves the submission (evaluator, not a build stakeholder)
(Illustrative, for blueprint realism) Bank AML Operations Head	Would own the business case in a real deployment	Referenced in blueprint governance/roadmap sections only — not a real contact for this capstone
(Illustrative) Bank Compliance / MLRO function	Would own escalation policy and final regulatory decisions in a real deployment	Referenced for the Human Decision boundary and escalation design only
(Illustrative) Bank IT / Data Owners	Would own real source systems in a real deployment	Referenced in Data Requirements as the target-state source of truth; capstone uses mock data instead

Note: The bank-side stakeholders above are illustrative — used to keep the blueprint realistic — not actual people the team has consulted. This is stated plainly so the document is not misread as claiming real institutional engagement.

3. RACI (who does what on this decision layer)

Decision	Responsible	Accountable	Consulted	Informed
Close a case with no findings	Investigator	Investigator	—	Team Lead
Escalate a case	Investigator	Investigator	Team Lead (optional)	Compliance
Accept escalation, decide further action	Compliance Officer	Compliance Officer	Investigator	Team Lead
Change trigger-rule thresholds	Admin	Team Lead	Investigators (feedback on false positives)	Compliance
Update the mock playbook	Admin	Team Lead	Investigators	—

This RACI illustrates the governance model the blueprint recommends (see **Section 6 — Technical Architecture** and blueprint governance section); the prototype simulates it with role-based views rather than real approval workflows.

4. Current Process / Workflow

Product: Sentinel AML — describes the **as-is** (today, without the product) process this capstone targets.

1. As-Is Process Narrative

1. A bank’s transaction-monitoring platform (existing system, out of scope to rebuild) applies rules across all transactions and raises an alert when a rule fires.
2. The alert lands in a case-management queue, typically with minimal context (customer ID, account ID, rule name).
3. An investigator is assigned or claims the case.
4. The investigator manually:
 - opens the core banking / CRM system to review customer and KYC details
 - opens the transaction system to pull recent history and compute (by eye or spreadsheet) whether the flagged activity is unusual
 - checks internal systems or spreadsheets for related accounts and counterparties
 - searches for prior alerts or cases involving the customer or counterparties
 - reviews any supporting documents already on file
 - manually writes an investigation summary, usually in a free-text case note or a Word/Excel template
5. The investigator decides: close (no concern), request more information, or escalate to a senior investigator / compliance.
6. The decision and rationale are recorded, typically as free text, in the case management system.
7. Audit trail exists only insofar as the case management system logs status changes; the evidence trail itself is reconstructed from notes, screenshots, and emails if ever revisited.

2. Pain Points (mapped to what Sentinel AML addresses)

Pain point	Effect	Addressed by
Evidence scattered across 4–6 systems	Time-consuming compilation before analysis can even start	AI Investigation Workspace pulls all evidence into one screen (mock data in prototype)
No structured way to say “I don’t have this information”	Gaps get missed or silently assumed away	Explicit Missing evidence-status category
Free-text write-ups vary investigator to investigator	Inconsistent quality, hard to review	Structured case report template, same fields every time
Hard to reconstruct “what was known when”	Weak defensibility on audit/regulatory review	Full audit log of AI actions and human decisions, timestamped
Investigators may treat every large transaction as suspicious (or the reverse — alert fatigue)	Both over- and under-investigation risk	Twin-case design (C1 suspicious / C2 legitimate) built into the product’s evidence logic — see Section 5 — Functional Requirements & Use-Case Document
No visible link between “why was this flagged” and “what should I check”	Investigator has to independently reconstruct the rule logic	Trigger Engine shows the exact rule and threshold that fired, attached to the case

3. Approvals, Exceptions, and Handoffs (as-is, and how the product maps to them)

Step	Who approves / handles today	Product equivalent
Alert generated	Automatic (existing monitoring system)	Trigger Engine (rule-based, F3)
Case assignment	Team lead / queue auto-assignment	Case queue dashboard (F2/F4)
Evidence gathering	Investigator, manual	AI Investigation Workspace (F5/F6)
Request for more information	Investigator emails/calls relevant teams	“Request information” action (F7) — logged, outcome manually re-entered by investigator (integration out of scope)
Escalation	Investigator flags to senior/compliance manually	Escalation queue (F9)
Final regulatory decision (e.g., filing)	Compliance/MLRO, outside the investigator’s system entirely	Out of scope — product stops at escalation hand-off

4. Future-State Summary (delta this product introduces)

The future-state process is identical in its decision authority (human investigator and compliance retain every consequential decision) but compresses steps 4–6 above from a multi-system manual exercise into a single AI-assembled workspace with mandatory citation and status tagging. See the two flowcharts:

- **Section 12 — Flowchart 1 – End-to-End Data Flow**
- **Section 13 — Flowchart 2 – Investigation Process Flow**

5. Functional Requirements & Use-Case Document

Product: Sentinel AML **Related:** Section 2 — Product Requirements Document (PRD)

1. Frozen Use Case

Scenario family: Sudden high-value credit followed by rapid movement of funds.

One-line statement: An AI assistant that turns an AML alert for rapid movement of a large credit into an evidence-backed investigation package, while the human investigator keeps the decision.

Two demonstration cases sharing the same trigger rules:

Case	Story	Outcome the demo shows
C1 — Suspicious	Customer with a small, steady transaction history suddenly receives ₹1 crore from an unrelated/unknown-relationship counterparty; ~86% moves out within 48 hours across 3 accounts (one hop external, thin data)	Multiple Verified red flags, several Missing items (no KYC on external hop accounts, undeclared source of funds), no counter-evidence on file
C2 — Legitimate twin	Same size and speed of movement, but the credit is a documented property-sale payment and the outbound transfer is to a verified family member, with a sale deed on file	Same trigger fires (amount deviation), but Verified counter-evidence is present; AI still lists the same category of red flag but with contrary evidence attached, and a lower recommended review priority — the human still decides

2. Alert Trigger Definitions

Rule ID	Name	Condition	Fires for
R1	Amount deviation	Single incoming credit $\geq 5\times$ the account's trailing 6-month average credit amount	C1, C2
R2	Rapid pass-through	$\geq 80\%$ of a flagged credit's value leaves the receiving account within 72 hours, across ≥ 2 outbound transactions/hops	C1, C2

Both rules must fire to create a case in the default demo configuration (reduces noise); thresholds are configurable by the Admin role (F10).

3. Step-by-Step Use Case: “Investigate a rapid-movement alert”

Actor: AML Investigator **Precondition:** Trigger Engine has raised an alert (R1 + R2 fired) and a case exists in the queue.

Step	User action	System response	Validation / error handling
1	Investigator opens case queue (landing page)	Displays open cases sorted by priority/age, shows alert type and rule(s) fired	Empty queue shows “No open cases” state
2	Investigator selects a case	Opens Investigation Workspace: case header, alert summary strip (trigger amount, baseline, review period)	If case data incomplete, shows explicit “Missing” banner rather than blank fields
3	Investigator reviews AI-generated findings	Findings panel shows red flags, counter-evidence, missing items, conflicts — each with evidence status and citations	Any finding without a citation is blocked by the grounding validator and never reaches this screen (see Section 6)
4	Investigator reviews transaction timeline & fund-flow view	Chronological chart + account/fund-flow graph render from the same underlying data the AI used	If a downstream account is external (no data), UI shows it as “External — no data available” rather than omitting it silently
5	Investigator reviews investigation questions & recommended next steps	Each next step cites a playbook entry ID	A next step with no playbook citation cannot be displayed (validator rule)
6	Investigator takes an action	Options: Request information / Add evidence / Escalate / Close	Escalate and Close require a non-empty rationale field before submission is allowed
7	System records the decision	Human Decision Panel updates; Audit Log receives a new entry with actor, action, timestamp, rationale	Decision is immutable once submitted (correction requires a new logged action, not an edit)

4. Expected Outcomes by Case

- **C1:** Investigator sees enough Verified + Missing evidence to justify escalation or a request for more information; decision remains theirs.
- **C2:** Investigator sees the same trigger but sufficient counter-evidence to justify closing with rationale “reviewed, documented legitimate transfer” — again, their decision, not the system’s.

5. Error / Edge Cases Covered

Case	Requirement
Missing data (stale KYC, no source-of-funds declared)	Shown as explicit “Missing” finding, never silently filled in or guessed
Conflicting data (declared occupation vs. observed activity mismatch)	Shown as “Conflicting” finding, both sources cited
Prompt injection via transaction reference text	Free-text fields are treated as untrusted data by the AI agent; validator strips/ignores any instruction-like content found there and logs the attempt
LLM/API unavailable during demo	Cached report replay available (F12) so the demo does not depend on a live call

6. Non-Goals for This Use Case

- No claim of criminal determination.
- No automatic filing of any regulatory report.
- No coverage of typologies outside the rapid-movement family in the Must-have scope (see Section 2 for Could-have extensions).

6. Technical Architecture

Product: Sentinel AML **See also:** Section 12 — Flowchart 1 – End-to-End Data Flow, Section 8 — Integration / API Specifications

1. Architecture Principle

Compute deterministically, reason with the LLM, verify before display.

- All numbers (baselines, totals, velocity, fund-flow chains) are computed by plain code, never by the LLM.
- The LLM only interprets computed results, drafts findings/questions/next steps, and must cite the record IDs it used.
- A grounding validator checks every AI output against the underlying data before it reaches the UI; unsupported claims are rejected or downgraded to “Inferred.”

2. System Layers

Layer	Responsibility	Technology (prototype)	Technology (target/production framing)
Data layer	Stores mock customer/account/transaction/relationship/alert/case/document/playbook records	CSV/Excel loaded into SQLite/DuckDB	Core banking, KYC system, case management DB, document store (real integrations — out of scope)
Monitoring layer	Rule-based Trigger Engine (R1, R2) scanning the transaction feed	Python scheduled/batch job over the mock feed	Real-time or near-real-time transaction monitoring platform (existing system in a real bank — not rebuilt here)
Tool layer	Deterministic functions: baseline calc, fund-flow trace, relationship lookup, prior-alert/case lookup, document lookup	Python functions, optionally wrapped as an MCP tool server	Same, backed by real source-system APIs
AI layer	Investigation Agent: selects tools, drafts structured report, cites sources	LLM with tool-calling + JSON-schema output	Same, with potential move to a multi-agent split (see phased design below)
Validation layer	Grounding validator: citation check, number check, forbidden-wording check	Python	Same, potentially with a second-model “critic” pass
Application layer	Case queue, Investigation Workspace, Human Decision Panel, Escalation Queue, Audit Log, Admin	Streamlit (single-tier, fastest path to a working prototype)	Web app: React/Next.js frontend + FastAPI (or similar) backend + auth service
Governance layer	Audit log, role-based views	Append-only log table + role selector (mock auth)	Immutable audit store, real RBAC/IAM, SIEM integration

Why Streamlit for the prototype, not a “real” web stack: given the team’s timeline (see Section 15), Streamlit gets a working, demoable product-shaped UI (multi-page, stateful, role-aware) built fastest. The architecture is still designed in layers so it maps cleanly onto a production frontend/backend split later — this is noted explicitly so evaluators see the target architecture, not just the shortcut.

3. Component Diagram (text form; see also the end-to-end flowchart)

```
Digital Channel (browser)
-> Sentinel AML App (Streamlit)
  -> Case Queue / Dashboard module
  -> Investigation Workspace module
  -> Human Decision module
  -> Escalation / Compliance module
  -> Admin module
-> Application talks to:
  -> Data Access Layer (SQLite/DuckDB over mock CSVs)
  -> Trigger Engine (rule evaluation service)
  -> AI Investigation Agent (LLM + tool layer)
  -> Grounding Validator
  -> Audit Log service
```

4. Why an Agent, Not a Chatbot

A chatbot answers one question at a time and puts the burden of asking the right questions on the user. The Investigation Agent instead runs a **fixed investigation procedure** (profile check → transaction/baseline analysis → fund-flow trace → relationship check → history check → evidence assembly → report), calling tools in that sequence, and always produces the same structured report shape. The user never has to “prompt it correctly” — this is what makes it a product feature, not a chat window bolted onto the case screen.

5. Phased Agent Design

- **Prototype (now):** one Investigation Agent orchestrating all tool calls.
- **Roadmap (documented, not built for the capstone):** split into six specialist agents — Alert Triage, Transaction Investigation, Relationship, Customer/KYC, Evidence, Investigation Summary — as originally proposed by the team. See **Section 15 — Delivery Plan / Milestone Roadmap**. The single-agent prototype is deliberately built so this split is additive, not a rewrite: each future agent takes over one existing tool-call sequence.

6. Security & Guardrail Notes (architecture-level; full list in Section 10 — Non-Functional Requirements)

- AI tools are **read-only** against the data store.
- Free-text fields (transaction reference, document text) are treated as **untrusted input**; the agent’s system prompt and the validator both guard against embedded instructions (prompt injection).
- No customer-identifying real data is used anywhere; the dataset is synthetic by construction (see **Section 7 — Data Requirements**).
- Every AI action and human decision writes to the audit log — this is treated as a non-optional side effect of the workflow, not a bolt-on.

7. Deployment (prototype)

Single machine or shared cloud dev environment running the Streamlit app and a local SQLite/DuckDB file; LLM calls go to a hosted API. No production deployment, container orchestration, or multi-environment setup is in scope for the capstone (see **Section 10 — Non-Functional Requirements** for what a production deployment would additionally require).

7. Data Requirements

Product: Sentinel AML — all data described here is **fictional**, generated by the team for demonstration only.

1. Core Entities

Entity	Purpose	Owner (target-state; prototype = team)	Retention (target-state, illustrative)
Customer	Profile, expected activity, risk rating, KYC status	KYC / onboarding system	Per applicable bank policy — not defined here
Account	Account master, internal/external flag	Core banking	As above
Transaction	Every credit/debit, channel, counterparty	Payments ledger	As above
Relationship	Known/unknown links between accounts	Relationship graph / observed	As above
Alert	Rule fired, trigger transaction, severity	Trigger Engine	As above
Rule	Rule ID, condition, threshold, version	Admin-configured	Versioned, never overwritten
PastCase	Prior investigation outcomes	Case management	As above
Document	Supporting evidence (e.g., sale deed)	Document store	As above
Playbook	Mock internal next-step guidance by scenario	Admin-authored (team)	Versioned
AIFinding	One row per AI-generated finding, with evidence status	Generated by AI layer	Immutable once written
AIReport	Full structured case report	Generated by AI layer	Immutable once written
HumanAction	Investigator/compliance decisions + rationale	User-entered	Immutable once written
AuditLog	Every system/AI/human action	System-generated	Immutable, append-only

2. Field-Level Detail

Customer

customer_id, name, type (individual/business), segment, occupation_or_industry, declared_annual_income, expected_monthly_credit, expected_monthly_debit, declared_source_of_funds, risk_rating, kyc_status, kyc_last_updated, country, onboarding_date

Account

account_id, customer_id (nullable — null for external accounts), type, open_date, currency, status, risk_rating, is_internal (boolean)

Transaction

txn_id, account_id, txn_datetime, direction (CR/DR), amount, currency, channel, counterparty_account_id, counterparty_name, reference_text, status

Relationship

relationship_id, account_from, account_to, type (family/employer/business/unknown), source (KYC-declared/observed), verified (Y/N/Unknown), start_date, previously_flagged

Alert

alert_id, case_id, customer_id, account_id, scenario_id, trigger_txn_id, rule_id(s), rule_version, alert_date, severity, status

Rule

rule_id, name, condition_description, threshold_value(s), version, effective_date, editable_by (Admin)

PastCase

case_id, alert_id, investigator, start_date, closure_date, disposition, rationale

Document

doc_id, customer_id, case_id, doc_type, doc_date, extracted_summary, verified (Y/N)

Playbook

playbook_id, scenario_id, step_no, step_text, escalation_criteria

AIFinding

finding_id, case_id, type (red_flag/counter_evidence/missing/conflicting), description, evidence_status, supporting_txn_ids, source_refs, why_it_matters, model_version, generated_at

AIReport

case_id, report_json, generated_at, model_version, validator_result

HumanAction

action_id, case_id, investigator, action, disposition, rationale (mandatory), findings_accepted, findings_rejected, timestamp

AuditLog

event_id, case_id, actor (AI/human/system), action, input_refs, output_refs, timestamp

3. Data Ownership (target-state mapping — prototype uses one team-owned mock dataset for all)

See Section 6 — Technical Architecture table (Data layer row) for the real-system equivalents this mock data stands in for.

4. Data Quality Requirements

- Every AIFinding.supporting_txn_ids / source_refs must resolve to an existing record — enforced by the grounding validator, not left to data hygiene alone.
- Deliberately **incomplete** records must exist in the dataset (e.g., no KYC on external accounts, missing declared_source_of_funds) so the “Missing” evidence status has real cases to surface — see test scenarios in Section 5 and Section 11.
- At least one deliberately **conflicting** pair of records must exist (e.g., declared occupation vs. observed transaction pattern) to exercise “Conflicting” status.

5. Privacy Needs

- No real personal data of any kind. All names, IDs, and amounts are synthetic.
- Even though fictional, the product itself must **behave** as if the data were sensitive (masked in any exported artifact, no data leaves the local environment, access is role-gated) — this is deliberate, so the demo also demonstrates privacy-appropriate handling as a design habit, not just a data-fictional convenience.

6. Sample Data Footprint

Table	Approx. rows for demo
Customer	8–10 (covers C1, C2, and 2–3 supporting cases from Section 5)
Account	15–20 (includes internal + external, chain accounts)
Transaction	150–300 (6–12 months history per active account, plus the alert-window transactions)
Relationship	10–15
Alert	6–8
Rule	2 (R1, R2) — 3 if the optional velocity rule is added
PastCase	3–4
Document	3–4

Table	Approx. rows for demo
Playbook	~10 rows covering the frozen scenario

7. Reporting Needs

- Case-level report (the AI-generated structured report) — primary “report” of this product.
- Audit log export (per case) for traceability review.
- No aggregate/BI reporting is in scope for the prototype; the roadmap document notes this as a future monitoring-dashboard capability.

8. Integration / API Specifications

Product: Sentinel AML **Scope note:** The prototype has **no external integrations** — everything runs against the mock dataset. This document specifies (a) the internal tool-layer “API” the AI agent actually calls, and (b) the external integrations a production version would need, clearly separated.

1. Internal Tool Layer (built and used in the prototype)

These are plain Python functions in the prototype; optionally exposed as MCP tools so the architecture matches the product story. Each has a fixed schema so the grounding validator can check every call.

Tool	Input	Output	Notes
<code>get_customer_profile(customer_id)</code>	<code>customer_id</code>	Customer record	Read-only
<code>get_account(account_id)</code>	<code>account_id</code>	Account record incl. <code>is_internal</code>	Read-only
<code>get_transaction_history(account_id, window)</code>	<code>account_id</code> , <code>date range</code>	List of transactions	Used for baseline calc
<code>compute_baseline(account_id)</code>	<code>account_id</code>	Trailing 6-month average/median credit and debit	Deterministic, code-only
<code>trace_fund_flow(txn_id, hops, time_window)</code>	<code>trigger txn_id</code> , <code>max hops</code> , <code>window</code>	Ordered chain of downstream transactions/accounts	Deterministic graph trace
<code>get_relationships(account_id)</code>	<code>account_id</code>	List of relationship records	Read-only
<code>get_prior_alerts_and_cases(customer_id_or_account_id)</code>	<code>id</code>	List of prior alerts/cases	Read-only
<code>get_documents(customer_id, case_id)</code>	<code>ids</code>	List of document summaries	Read-only
<code>get_playbook(scenario_id)</code>	<code>scenario_id</code>	Ordered list of next-step entries	Used to ground “next steps”
<code>evaluate_rules(transaction_feed_window)</code>	<code>feed window</code>	List of fired rule+alert candidates	Trigger Engine only — not called by the AI agent

Authentication: none required in-prototype (single local process). **Failure handling:** any tool call that raises an error is caught, logged to AuditLog with `actor=system`, and surfaced to the agent as an explicit “data unavailable” result rather than silently retried or guessed around.

2. External Integrations — Target State Only (NOT built for this capstone)

System	Purpose	Likely protocol	Notes
Core banking / transaction ledger	Real transaction feed	Batch file or streaming API (bank-specific)	Replaces mock Transaction table
KYC / CKYC / PAN verification	Real customer verification	REST API, licensed integration	Replaces mock Customer.kyc_status
Credit/AML screening providers	Sanctions/PEP/adverse media (if the product scope later expands to screening)	REST API, licensed integration	Not part of the frozen use case; noted for roadmap only
Case management system	System of record for case status	REST API or DB integration	Prototype’s case queue stands in for this
Document management system	Real supporting documents	REST API / file store	Replaces mock Document table
LLM provider	Investigation Agent reasoning + report generation	REST API (chat/completions with tool-calling, JSON-schema output)	This is the one real external call the prototype makes — see below

3. The One Real External Call: LLM API

```

+-----+
| CASE-001 | Customer CUST-004 | Account ACC-1004 | Status: Open |
| Alert trigger: Rule R1+R2 fired - 50x baseline, 86% moved in 48h |
+-----+
| ALERT SUMMARY STRIP |
| Trigger amt: Rs 1,00,00,000 | Baseline avg: Rs 2,00,000 |
| Txns in window: 4 | Accounts involved: 4 | Review period: 72h |
+-----+
| TRANSACTION TIMELINE | FUND-FLOW / NETWORK VIEW |
| Sep20 +1,00,00,000 (credit) | CUST-004 |
| Sep20 -95,00,000 -> ACC-B | |
| Sep21 -90,00,000 -> ACC-C | ACC-A --95L--> ACC-B |
| Sep21 -85,00,000 -> ACC-D (external) | --90L--> ACC-C |
| | --85L--> ACC-D (external, no data) |

```

```
+-----+
| AI FINDINGS (AI recommendation - draft) |
| [VERIFIED] Amount 50x historical average [cite: TXN-10543] |
| [VERIFIED] 86% moved out within 48h [cite: TXN-...] |
| [MISSING] No KYC on record for ACC-D (external) |
| [MISSING] No declared source of funds for the credit |
+-----+
| INVESTIGATION QUESTIONS | RECOMMENDED NEXT STEPS |
| - Where did the RslCr come from?| 1. Request source-of-funds doc |
| - Is ACC-B/C/D previously flagged| 2. Review relationship ACC-A/B |
| | [playbook: PB-AML-07-02] |
+-----+
| HUMAN DECISION (separate panel, distinct color/border) |
| Action: ( ) Request info ( ) Add evidence ( ) Escalate ( ) Close |
| Rationale (required): [_____] |
| Findings accepted: [x][x][ ][x] Findings rejected: [ ] |
| | [ Submit decision ] |
+-----+
| Status: PENDING HUMAN REVIEW |
+-----+
```

Design note: the AI Findings panel and the Human Decision panel use visibly different colors/borders (e.g., AI = blue outline “draft,” Human = green outline “action required”) so a first-time viewer instantly sees the boundary the product enforces.

Screen 4 — Escalation / Compliance Queue

```
+-----+
| COMPLIANCE REVIEW QUEUE |
| Case Escalated by Reason Date |
| CASE-001 Investigator A Unverified source of funds Sep22 |
+-----+
| [Open case - read-only investigation view + escalation note] |
+-----+
```

Screen 5 — Audit Log Viewer

```
+-----+
| AUDIT LOG - CASE-001 |
| Time Actor Action |
| 09:02:11 System Alert generated (R1+R2, rule v1.0) |
| 09:02:12 System Case created, added to queue |
| 09:14:03 AI Agent Called get_transaction_history |
| 09:14:05 AI Agent Called trace_fund_flow |
| 09:14:22 AI Agent Report generated, validator: PASS |
| 10:01:07 Investigator Reviewed case |
| 10:12:40 Investigator Action: Escalate - rationale: "... " |
+-----+
```

Screen 6 — Admin / Rule Configuration (light)

```
+-----+
| RULE CONFIGURATION |
| R1 Amount deviation Threshold: [5]x baseline [Save] |
| R2 Rapid pass-through Threshold: [80]% within [72] hours [Save] |
+-----+
| PLAYBOOK ENTRIES (scenario: rapid-movement) |
| PB-AML-07-01 Review customer KYC/profile information |
| PB-AML-07-02 Review relationship between sending/receiving accts |
| ... [Edit] |
+-----+
```

Mobile / Responsive Requirements

Not required for the capstone demo (desktop-only, presented on a laptop/projector). Noted as a target-state requirement in **Section 10 — Non-Functional Requirements** rather than built.

Accessibility Expectations

- Evidence-status tags use **color + text label** together (not color alone), since color-only status indicators fail accessibility and are also hard to read on a projector.
- All screens keep a persistent text banner: “Prototype — fictional data. AI assists; the investigator decides.” (not relying on a tooltip or icon alone).

10. Non-Functional Requirements

Product: Sentinel AML **Note:** Split into **Prototype (built)** and **Target-state (documented for the blueprint, not built)** — evaluators should be able to see the team understands the gap, not have it hidden.

1. Performance

Requirement	Prototype	Target-state
Case report generation time	Cached instantly; live generation a few seconds, acceptable for a demo	Sub-second for cached/common cases; SLA-defined for live generation at scale
Concurrent users	1 (demo machine)	Full investigation team, concurrent access with no degradation

2. Availability

Requirement	Prototype	Target-state
Uptime	Best-effort, local/dev environment	Defined SLA (e.g., business-hours or 24/7 depending on bank operating model) with monitoring and on-call

3. Accessibility

- **Prototype:** color + text labels together for evidence status (see Section 9); readable font sizes for projector demo.
- **Target-state:** WCAG 2.1 AA compliance across the application.

4. Audit History

- **Prototype (built):** every AI tool call and every human decision writes an immutable AuditLog row; log is viewable per case (Screen 5).
- **Target-state:** tamper-evident storage, retention aligned to regulatory requirements, SIEM integration, log-access itself audited.

5. Security

Requirement	Prototype	Target-state
Data in transit	LLM API call over HTTPS (provider default)	End-to-end encryption, VPN/private link to core systems
Data at rest	Local file/SQLite, no encryption (data is fictional)	Encryption at rest, key management, data residency controls
Authentication	Mock role selector, no real credentials	Full IAM/SSO integration, MFA
Authorization	Role-based screen visibility only (soft enforcement)	Enforced RBAC at the API layer, least-privilege
Prompt injection	Free-text fields treated as untrusted; validator strips instruction-like content; tested explicitly (see Section 5 edge cases)	Same principle, hardened and continuously red-teamed
Secrets management	API key kept out of shared repo/screen-share, held by one team member	Vault/secrets-manager, rotated keys, no individual ownership

6. Roles

- **Prototype:** four roles simulated via a selector at login (Investigator, Team Lead, Compliance, Admin) — see **Section 3 — User Personas & Stakeholders**.
- **Target-state:** full RBAC tied to HR/identity systems, least-privilege data access per role (e.g., Compliance sees escalated cases only, Admin cannot see case content, only configuration).

7. Compliance (as a design habit, not a legal claim)

- The product **never** asserts a legal/regulatory conclusion (BR1).
- No specific regulation is cited by the AI output; any regulatory reference used in the blueprint document itself is flagged for verification by a team member with compliance/banking background (see **Section 14 — Risks, Assumptions, Dependencies & Constraints**).
- Target-state would require formal model-risk governance, a defined AML program owner, and legal/compliance sign-off before any production use — explicitly out of scope for an academic capstone.

8. Scale

Requirement	Prototype	Target-state
Data volume	~8–10 customers, ~150–300 transactions (see Section 7)	Full transaction volume of a bank’s monitored population
Case volume	6–8 demo cases	Full alert volume with prioritization/queuing at scale

9. Explainability

- **Prototype (built):** every AI finding carries an evidence status and a citation; the grounding validator is the explainability enforcement mechanism, not just a UI label.
- **Target-state:** formal model documentation, periodic explainability review as part of model risk management.

10. Reliability of the Demo Itself (a prototype-specific NFR)

- Cached report replay (F12) so a live-API failure does not derail the 15-minute presentation.
- Pre-validated fixed dataset (no random generation at runtime) so results are reproducible on rehearsal and on demo day.

11. Acceptance Criteria / Definition of Done

Product: Sentinel AML

1. Feature-Level Acceptance Criteria

F3 — Trigger Engine

- ☐ Given the mock transaction feed, when a credit meets Rule R1 ($\geq 5\times$ baseline) AND a subsequent Rule R2 ($\geq 80\%$ out within 72h across ≥ 2 hops), an alert is created with the exact rule ID(s) and threshold values attached.
- ☐ Given a transaction that meets only R1 but not R2 (e.g., C7 in test scenarios), no alert is created in default config.
- ☐ The rule(s) and threshold values that fired are visible on the case (not just “alert exists”).

F5/F6 — Investigation Workspace & Evidence Status

- ☐ Every AI finding displayed has a non-empty citation (`supporting_txn_ids` or `source_refs`).
- ☐ Every AI finding has exactly one evidence status: Verified, Inferred, Missing, or Conflicting.
- ☐ A finding is never labeled “Verified” unless the validator confirms the cited record directly supports the stated claim.
- ☐ For case C1, at least one “Missing” finding appears (external-account KYC gap).
- ☐ For case C2, at least one “Conflicting” or counter-evidence finding appears that a reviewer can see differs meaningfully from C1’s findings for the same trigger.

F7 — Human Decision Panel

- ☐ Escalate and Close actions are blocked until a non-empty rationale is entered.
- ☐ Once submitted, a decision cannot be silently edited — a correction requires a new logged action.
- ☐ The Human Decision Panel is visually distinct from the AI Findings panel at a glance (color/border difference).

F8 — Audit Log

- ☐ Every AI tool call during a case’s investigation appears as a log entry.
- ☐ Every human action appears as a log entry with actor, timestamp, and rationale (where applicable).
- ☐ Log entries cannot be edited or deleted through the UI.

F12 — Cached/Live Toggle

- ☐ With no network/API access, the demo cases still load a full report from cache.
- ☐ Switching to “live” regenerates a report for at least one case without error, when API access is available.

2. Guardrail Acceptance Criteria (from BR1–BR8)

- ☐ No AI-generated text anywhere in the product uses the words “criminal,” “illegal,” “guilty,” or equivalent verdict language (automated check + manual review).
- ☐ No AI-generated “next step” appears without a `playbook_id` citation.
- ☐ The prompt-injection test case (C8 in Section 5) produces a report that does not follow the injected instruction, and the attempt is visible in the audit log.

- ☐ Every screen carries the “Prototype — fictional data” banner.

3. Definition of Done — Prototype

The prototype is **Done** for submission when:

1. All Must-have features (F1–F8, F11) from **Section 2 — Product Requirements Document (PRD)** work end to end for both C1 and C2 without manual intervention.
2. All acceptance criteria in sections 1–2 above pass.
3. The full flow — trigger fires → case in queue → investigation workspace with cited findings → human decision recorded with rationale → audit log entry visible — has been run start to finish at least twice successfully (once in rehearsal, once on demo hardware) per **Section 15 — Delivery Plan / Milestone Roadmap**.
4. A cached fallback exists for every demo case in case of live API failure.

4. Definition of Done — Documentation Pack

1. All 14 documents in this pack are internally consistent (same product name, scenario names, rule IDs across all files), and cross-reference each other correctly.
2. Both flowcharts render correctly when pasted into a Mermaid-compatible tool.
3. The full pack has been reviewed against the sample blueprint’s evaluation rubric by at least one team member other than the author of each section.

5. Definition of Done — Presentation

1. Slide deck built from this documentation pack (not duplicating it verbatim).
2. One presenter identified and rehearsed within the 15-minute limit including Q&A buffer.
3. Live demo (or a screen recording, as a fallback) of the end-to-end flow is ready.

12. Flowchart 1 – End-to-End Data Flow

Product: Sentinel AML **How to use this section:** Copy the code block below into mermaid.live, the Mermaid VS Code extension, or draw.io’s “Insert > Mermaid” option to get a rendered, editable diagram for the slide deck (this PDF renders it as text, not a picture).

Shows: how a transaction becomes a trigger, how a trigger becomes a case, how the AI agent assembles evidence, and how the case reaches a human decision and the audit log.

```
graph TD
    subgraph SRC ["Mock Bank Systems (fictional data)"]
        TXN[Transaction Feed]
        CUST[Customer / KYC Store]
        ACCT[Account Master]
        REL[Relationship / Counterparty Store]
        HIST[Prior Alerts and Cases]
        DOC[Document Store]
    end

    TXN --> TE
    CUST --> TE
    ACCT --> TE

    subgraph MON ["Monitoring Layer - rule-based, no AI"]
        TE["Trigger Engine, Rule R1: amount deviation, Rule R2: rapid pass-through"]
    end

    TE -->|Alert generated| ALERT["Alert Record, rule id + thresholds"]
    ALERT --> CASE["Case Created, added to Case Queue"]

    CASE --> DASH["Investigator Dashboard, Case Queue - landing page"]
    DASH --> CLAIM["Investigator claims case"]

    CLAIM --> AGENT

    subgraph AI ["AI Investigation Layer"]
        AGENT["AI Investigation Agent"]
        TOOLS["Deterministic Tools: baseline calc, fund-flow trace, relationship lookup, history lookup"]
        AGENT <--> TOOLS
    end

    CUST --> TOOLS
    ACCT --> TOOLS
    REL --> TOOLS
```



```

HIST --> TOOLS
DOC --> TOOLS
TXN --> TOOLS

AGENT --> VALID["Grounding Validator, checks citations and numbers"]
VALID --> REPORT(["AI Case Report: red flags, evidence status, questions, next steps"])

REPORT --> WS["Investigation Workspace, Investigator reviews"]
WS --> DECISION{"Human Decision"}
DECISION -->|Close| CLOSE["Case Closed, + rationale"]
DECISION -->|Request info| REQ["Request Information, back to investigation"]
DECISION -->|Escalate| ESC["Compliance Review Queue"]

CLOSE --> AUDIT
REQ --> AUDIT
ESC --> AUDIT
ESC --> COMP["Compliance Officer Review"]
COMP --> AUDIT

subgraph GOV["Audit and Governance"]
  AUDIT[("Immutable Audit Log")]
end

```

Reading Notes for the Blueprint / Deck

- The **Monitoring Layer** is deliberately rules-only — no AI/LLM involvement in raising the alert. This is worth calling out on the slide: it shows the team understood not every step should use GenAI (see Section 6 §1, §4).
- The **AI Investigation Layer** never writes back to source data — tools are read-only (see Section 8 §1).
- Every path out of “Human Decision” ends in the **Audit and Governance** block — this is the diagram’s visual proof of the human-in-the-loop guarantee.
- Cross-reference: Section 13 zooms into the “Investigation Workspace -> Human Decision” portion of this diagram.

13. Flowchart 2 – Investigation Process Flow

Product: Sentinel AML **How to use this section:** Copy the code block below into mermaid.live, the Mermaid VS Code extension, or draw.io’s “Insert > Mermaid” option to get a rendered, editable diagram.

Shows: the investigator’s journey through a single case, once it has landed in the queue — this is the zoomed-in view of the “Investigation Workspace -> Human Decision” part of Section 12.

```

flowchart TD
  START(["Case appears in queue"]) --> CLAIM["Investigator claims / opens case"]
  CLAIM --> HEADER["Review case header: alert reason, customer, account, trigger"]
  HEADER --> AIPANEL["Open AI-generated evidence pack"]

  AIPANEL --> P1["Customer and KYC Review, profile vs expected activity"]
  AIPANEL --> P2["Transaction and Fund-Flow Review, timeline, baseline deviation"]
  AIPANEL --> P3["Relationship / Network Review, counterparties, connections"]
  AIPANEL --> P4["Prior Alerts and Case History Review"]

  P1 --> FLAGS
  P2 --> FLAGS
  P3 --> FLAGS
  P4 --> FLAGS

  FLAGS["Review Red Flags and Evidence, each tagged Verified / Inferred / Missing / Conflicting"]
  FLAGS --> QCHECK{"Enough evidence to decide?"}

  QCHECK -->|No - info missing| REQINFO["Request additional information or documents"]
  REQINFO --> WAIT["Await response / add evidence"]
  WAIT --> FLAGS

  QCHECK -->|Yes| QUESTIONS["Review AI investigation questions and recommended next steps (cited from playbook)"]
  QUESTIONS --> JUDGE["Investigator applies judgment: accept / reject each AI finding"]

  JUDGE --> DECIDE{"Disposition decision"}
  DECIDE -->|No concern found| CLOSE_OK["Close case - no action, + written rationale"]
  DECIDE -->|Needs senior / compliance review| ESCALATE["Escalate to Compliance / Senior Investigator, + written rationale"]
  DECIDE -->|Still unclear| MORE["Keep open - continue investigation"]

  CLOSE_OK --> LOG["Record decision and rationale in Audit Log"]
  ESCALATE --> LOG
  MORE --> WAIT

  LOG --> END(["Case updated / closed in system of record"])

```

Reading Notes for the Blueprint / Deck

- The loop **FLAGS -> QCHECK -> REQINFO -> WAIT -> FLAGS** is the “Missing evidence” path — it shows the product doesn’t force a decision when information is genuinely absent (see the Missing evidence-status requirement in Section 5 and Section 11).
- **JUDGE** (“accept / reject each AI finding”) is the step that makes the human-in-the-loop real rather than nominal — the investigator isn’t just clicking “approve,” they are individually accepting or rejecting each finding.
- The three colored outcomes (green = close, red = escalate, yellow = keep open) map directly to the Human Decision Panel actions in Section 9, Screen 3.
- Walking through this diagram live with case **C1** (ends in Escalate) and then case **C2** (ends in Close) on the same rule trigger is the single strongest 5 minutes of the demo — see Section 5 §1.

14. Risks, Assumptions, Dependencies & Constraints

Product: Sentinel AML

1. Assumptions (stated explicitly so they are never mistaken for validated facts)

#	Assumption	Where it’s used	Needs validation before reuse outside this capstone
A1	Investigators currently spend “several hours” compiling evidence per moderately complex case	Section 1	Yes — illustrative only
A2	6× reduction / “minutes instead of hours” style time savings	Value story, ROI language	Yes — no measured baseline exists
A3	India (PMLA/FIU-IND/RBI) is an acceptable illustrative jurisdiction context	Scenario framing	Team decision pending; no specific regulation cited without a compliance-literate reviewer’s check
A4	An LLM with tool-calling and structured JSON output is accessible to the team at low/no cost	Section 8	Yes — provider/key not yet assigned (see Section 16)
A5	Streamlit is an acceptable prototype UI technology for the evaluation	Section 6	Team preference; document explicitly frames it as a stand-in for a target web architecture
A6	7-person team, ~10 working days of effective build time	Section 1, Section 15	Confirm actual availability per person

2. Risks

#	Risk	Likelihood	Impact	Mitigation
R1	Live LLM call fails or behaves unexpectedly during the 15-minute demo	Medium	High	Cached report replay (F12); rehearse with network off at least once
R2	Mock data looks too clean/scripted, undermining credibility	Medium	Medium	Deliberately add irregular timing, a legitimate-looking small outflow, and the C2 twin case (see Section 5 §2)
R3	Team runs out of time before all Must-have features are built	Medium	High	Strict MoSCoW prioritization in Section 2; Should/Could items dropped first, not Must items
R4	Regulatory references added late by someone without compliance background, introducing inaccurate claims	Medium	High	Named reviewer requirement logged in Section 16; default to no specific-regulation claims if unverified
R5	Evaluators perceive the product as “AI decides,” undermining the human-in-the-loop positioning	Low–Medium	High	Enforced via UI design (visually separate panels), BR1/BR4 business rules, and explicit acceptance criteria in Section 11
R6	LLM produces an unsupported claim (“hallucination”) in a finding	Medium	High	Grounding validator rejects/downgrades any finding without a citation (see Section 6 §1)
R7	Prompt injection via transaction reference or document text	Low–Medium	Medium	Explicit test case C8 in Section 5; untrusted-input handling in validator
R8	Team members duplicate work or edit the same section simultaneously	Medium	Low–Medium	Single-owner-per-workstream model in the team-plan workbook; one master document owner
R9	No confirmed LLM API access close to the build deadline	Medium	High	Assign an owner immediately (open item in Section 16); have a rules-only fallback demo path (no live LLM, cached only) as a last resort

3. Dependencies

Dependency	Needed for	Status
LLM API access (provider + key)	AI Investigation Agent, live demo mode	Not yet assigned — see Section 16
Team availability per the timeline	All build workstreams	Assumed per A6; not yet individually confirmed
Confirmed exact submission date/format	Final scheduling of all deadlines in Section 15	Not yet confirmed — email only gives “1st/2nd week of October”
A team member with banking/compliance familiarity	Reviewing any regulatory language before submission	Not yet assigned

4. Constraints

Constraint	Detail
No real data	Fictional data only, by design and by necessity (no institutional access)
No budget	Academic project; must use free/low-cost tools throughout
Timeline	Submission window in the first/second week of October 2026; today is 27 September 2026
Team size and skill mix	7 members with unconfirmed individual strengths (see Section 3 and the team-plan workbook)
No production access or certification	Explicitly out of scope; documented as target-state only throughout this pack

5. How This Log Should Be Used

Review and update this log at each team check-in alongside Section 16. An assumption that gets validated (or a risk that materializes) should be noted with a date, not silently removed.

15. Delivery Plan / Milestone Roadmap

Product: Sentinel AML **Note:** Dates are proposals working backward from “1st/2nd week of October” — confirm the exact submission date and adjust (see Section 16, open item).

1. Capstone Delivery Milestones (this submission)

Date (proposed)	Milestone	Depends on
27–28 Sep 2026	Use case, jurisdiction, and track confirmed by team; workstreams claimed	Team call
29 Sep 2026	Mock dataset spec finalized (customers, accounts, transactions for C1–C4 minimum)	Use case confirmed
30 Sep – 1 Oct	Trigger Engine (R1, R2) + data layer built and tested against the dataset	Dataset ready
1–2 Oct	AI Investigation Agent + tool layer + grounding validator built	Trigger Engine, LLM access confirmed
2–3 Oct	Investigation Workspace UI built (case queue, workspace, decision panel)	Agent producing valid reports
3 Oct	Audit log, escalation queue, admin screen added	Core workspace working
4 Oct	Full end-to-end run-through: trigger → case → AI report → decision → audit, for C1 and C2	All modules integrated
4–5 Oct	Cached report fallback captured for demo reliability; test scenarios C3–C8 run	End-to-end working
5 Oct	Documentation pack finalized and cross-checked	Prototype behavior matches documents
5–6 Oct	Slide deck built; presenter rehearsal (timed, 15 minutes incl. Q&A)	Documentation + prototype done
6 Oct	Internal freeze — final QA pass against Section 11	Everything above
1st–2nd week Oct	Submission (exact date TBD)	Freeze complete

2. Phased Roadmap Beyond the Capstone (target-state — documented for blueprint completeness, not built)

Phase	Objective	Key activities	Not part of this capstone
Phase 1 — Real data discovery	Validate assumptions in Section 14 against a real institution	Process mapping, data inventory, measured baseline (actual hours/alert, false-positive rate)	Entire phase
Phase 2 — Real integrations	Replace mock data with real source systems	Core banking, KYC, case-management integration (see Section 8 target-state table)	Entire phase
Phase 3 — Multi-agent split	Move from single Investigation Agent to the six-agent design (Triage, Transaction, Relationship, KYC, Evidence, Summary)	Agent-by-agent migration, each replacing one tool-call sequence	Documented design only (see Section 6 §5)

Phase	Objective	Key activities	Not part of this capstone
Phase 4 — Screening & policy RAG	Add sanctions/PEP/adverse-media review and real policy-document retrieval	Vendor selection, RAG index over approved policy documents	Out of scope; noted as extension
Phase 5 — Governance & model risk	Formal model risk management, bias/drift monitoring, compliance sign-off	Model documentation, monitoring dashboards, periodic review	Out of scope
Phase 6 — Production hardening	Full IAM, encryption, scale testing, SLAs	Security certification, load testing	Out of scope
Phase 7 — Pilot and scale	Controlled pilot with real (masked) data, then broader rollout	Pilot design, adoption plan, KPI baselining	Out of scope

3. KPIs (target-state, illustrative — see Section 14, A1/A2 for the assumption flag)

KPI	How it would be measured in production	Not measured in this prototype
Time per alert (compile-to-decision)	Timestamp diff, case creation to disposition	Yes — prototype has no real user population
False-positive rate after AI-assisted triage	Investigator disposition vs. alert volume	Yes
Evidence completeness at first review	% of findings with no “Missing” status at case open	Partially observable in demo cases, not a measured population metric
Investigator satisfaction / adoption	Survey, usage metrics	Yes
Audit completeness	% of decisions with full traceable evidence chain	This one the prototype can actually demonstrate structurally — every decision in the prototype has a full chain by construction

4. ROI Logic (illustrative, explicitly marked as assumption-based)

A simple illustrative frame for the blueprint’s value section — **not a claim of actual measured savings**:

If an investigator currently spends an assumed *N* hours compiling evidence per case, and Sentinel AML reduces that to *M* minutes of review time, the illustrative time saved per case is (*N* hours – *M* minutes), which can be multiplied by case volume and loaded cost per investigator-hour to estimate an illustrative annual saving. **Both N and M are placeholders for the team to fill in only if they choose to include this in the blueprint, and must be labeled as assumptions, not measured results.**

5. Final Recommendation & Decisions Needed (for the blueprint’s closing section)

1. Confirm the twin-case (C1/C2) design as the anchor demo — recommended.
2. Confirm Blueprint + shallow prototype as the submission track — recommended, given the timeline.
3. Assign the open items in Section 16 before Phase-1-equivalent build work starts (29 Sep target).
4. Treat everything in section 2 of this document as the “roadmap beyond the capstone” slide — it demonstrates forward thinking without implying it was built.

16. Meeting Notes and Decisions (Open Items Log)

Product: Sentinel AML **Purpose:** Running log of decisions, unresolved questions, action owners, and deadlines. **Every item below is a proposal or an open question awaiting human confirmation from the team — nothing here is final until the team confirms it on a call or in writing.**

Log Format

Each entry: Date | Topic | Decision or Open Question | Owner | Status

Entries

Date	Topic	Decision / Open Question	Owner	Status
2026-09-26	Use case	Proposed: “sudden high-value credit followed by rapid movement of funds,” with a suspicious case (C1) and a legitimate twin case (C2)	Team	Needs team confirmation

Date	Topic	Decision / Open Question	Owner	Status
2026-09-26	Jurisdiction	Proposed: India context (illustrative only; no specific regulation cited without verification)	Team	Needs team confirmation
2026-09-26	Submission track	Proposed: Blueprint + shallow prototype	Sweta (stated preference)	Needs final team confirmation
2026-09-27	Trigger mechanism	Decision direction: build a visible, rule-based Trigger Engine (R1 amount deviation, R2 rapid pass-through) as a real first stage of the demo, not a pre-seeded alert	Team	Needs team confirmation
2026-09-27	Product framing	Decision direction: build as an end-to-end product (landing page, case queue, investigation workspace, human decision panel, audit log) rather than a single chatbot screen	Team (per Sweta's direction)	Adopted for this documentation pack — confirm at next call
—	LLM provider & API key/budget	Open. Which provider (e.g., a specific vendor), whose account, who pays for/monitors usage?	Unassigned — needs owner	Open
—	Presenter	Open. Who is the one presenter for the 15-minute slot?	Unassigned	Open
—	Master document ownership	Open. Who merges all sections into the final submission and does final QA? Suggested: Sweta, per earlier team-plan draft	Unassigned — needs confirmation	Open
—	Workstream owners	Open. See the team-plan workbook (<i>Capstone Team Plan.xlsx</i>) — owners not yet assigned to workstreams A–J	Unassigned	Open
—	Exact submission date/portal	Open. Email says “1st/2nd week of October” — exact date and submission mechanism not yet confirmed	Unassigned	Open
—	Second trigger rule (velocity/structuring, F13)	Open. Build only if time allows after Must-haves are done	Team	Open — deprioritized by default
—	Regulatory references in blueprint	Open. Any specific PMLA/FIU-IND/RBI citation added to the blueprint text must be checked by a team member with compliance/banking background before submission	Needs a named reviewer	Open

Unresolved Questions Needing a Team Decision

1. Do we confirm the use case and twin-case design as final, or does anyone have a competing AML scenario to propose?
2. Is the India/PMLA framing acceptable, or should the team stay jurisdiction-agnostic to avoid citing anything unverified?
3. Who owns the LLM API key and cost, and what is the budget ceiling (even if near-zero)?
4. Who is the single presenter, and who is the backup if that person is unavailable on presentation day?
5. Confirm: Streamlit for the prototype UI (fastest path), or does anyone want to push for a React/FastAPI build given more lead time?

Action Items (until claimed, these have no owner)

Action	Suggested owner	Deadline (proposed)
Confirm use case, twin case, jurisdiction on next team call	All	Next call
Claim workstreams A–J in team-plan workbook	All	Within 24h of next call
Provide/confirm LLM API access	TBD	Before prototype build starts
Verify any regulatory references before they go in the blueprint	TBD (compliance/banking background preferred)	Before final submission
Confirm exact submission date and format from IIT Mumbai	Sweta or admin contact	ASAP

Reminder: This log should be updated after every team call, not rewritten from scratch — add new rows, don't delete history.